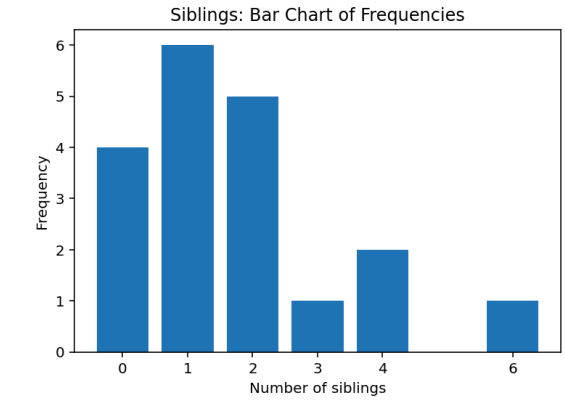
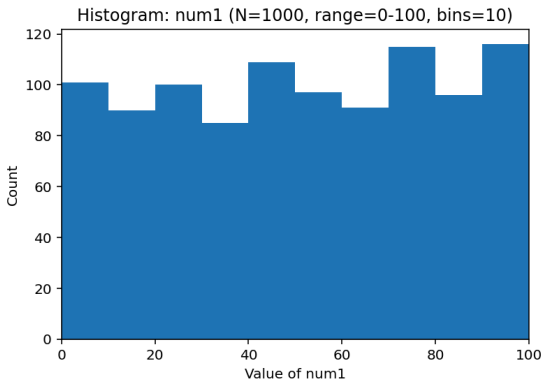
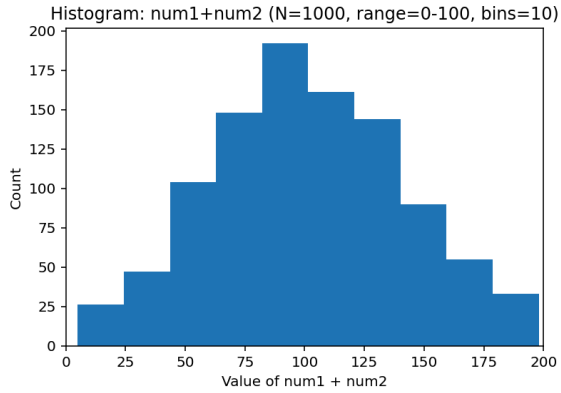
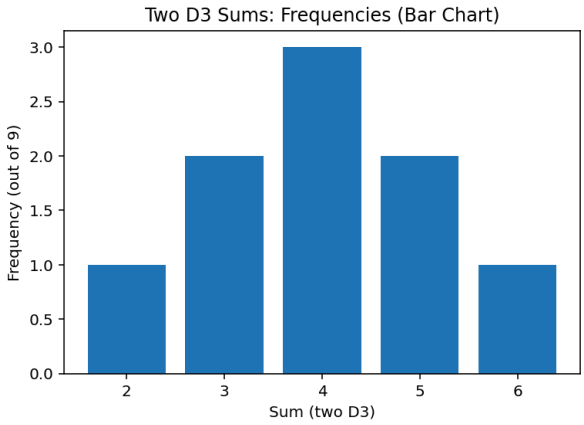
Plotting allows the reader to more easily assess what the data looks like, whether it looks normally distributed, bi-modal, skewed, etc. Conventionally, different types of plots are used to visualize different types of data.

Graphs:









Part 1:
The dataset was the number of siblings for a group of people. From the bar chart, the most common value is 1 sibling. The next most common is 2 siblings. Values like 3, 4, and 6 appear much less often. Most of the data is on the lower end (0, 1, 2), and then it drops off as the number gets bigger. Because of that, the distribution is not normal. It is discrete data and looks right-skewed since there are a few larger values stretching out to the right. The bar chart works well here because these are countable categories.

Part 2:
The histogram of num1 looks mostly flat across the range from 0 to 100. The bars are not exactly equal, but they are roughly the same height. This makes sense because every number in the range has about the same chance of being chosen. So this is like a uniform distribution.

The histogram of num1 + num2 looks very different. It forms a hill shape, with the highest bars in the middle and lower bars at the edges. There are many ways to get middle values (like around 100), but only a few ways to get very small or very large sums. Because of that, num1 + num2 is closer to a normal (bell-shaped) distribution than num1 alone.

Effect of changing the range:
When the range of numbers is larger, there are more possible values. The histogram spreads out more, but the sum still peaks in the middle. When the range is smaller, the histogram looks more blocky because there are fewer possible outcomes.

Effect of changing the sample size:
When the number of samples increases, the histogram becomes smoother and the shape is clearer. Random ups and downs even out. When the sample size is small, the histogram looks more uneven and noisy.

Effect of changing the number of bins:
Using more bins shows more detail, but the graph can look more jagged. Using fewer bins makes the graph look smoother, but it hides some detail. Changing bins mostly changes how the graph looks, not the actual data.

Kurtosis, skew, and modality:
Changing the number of bins does not actually change kurtosis, skew, or modality. It only changes the visual appearance. Changing the sample size also does not change the true distribution, but with small samples it might look more skewed just by chance. Changing the range does change what values are possible, so it can change the spread and overall shape. The num1 graph looks uniform and unimodal. The num1 + num2 graph is unimodal and more symmetric with a central peak.

Part 3:
For one D3 roll, the possible outcomes are 1, 2, and 3. Each one has probability 1/3. For two D3 rolled and added, the possible sums are 2, 3, 4, 5, and 6. From the bar chart, the frequencies are:
2 → 1 way
3 → 2 ways
4 → 3 ways
5 → 2 ways
6 → 1 way
So the probabilities are 1/9, 2/9, 3/9, 2/9, and 1/9. The shape is like a triangle with the highest point at 4. This matches the idea that middle values are more likely than extremes.

Comparison to Part 2:
The two D3 sum bar charts look similar to the histogram of num1 + num2. Both have a peak in the middle and fewer extreme values. The single D3 roll (which would be uniform) is similar to the num1 histogram, which is flat. This shows how adding random values tends to create a distribution that is more centered and bell-shaped.